5 Lara is a skier.

She completed a ski race in 1 minute 54 seconds.
The race was 475 m in length.

Lara assumes that her average speed is the same for each race.

(a) Using this assumption, work out how long Lara should take to complete a 700 m race.
Give your answer in minutes and seconds.

1 min 54 s = 60 + 54 = 114 s

	Metres	Seconds	
÷19	475	114	÷19
×28	25	6	×28
	700	168	

1 mark
Use one consistent time unit and scale.

1 mark
Reach 700 m using the same speed.

168 s = 120 s + 48 s = 2 min 48 s

1 mark
Correct time.

2 min 48 s
..... minutes ..... seconds

(3)

Lara's average speed actually increases the further she goes.

(b) How does this affect your answer to part (a)?

She would take less time, so my estimate

of 2 min 48 s would be too high.

(1)

1 mark
Explain the effect on the time estimate.

(Total for Question 5 is 4 marks)